

Linear Programming Report

Sensitivity Analysis & Duality

Infeasible Primal / Infeasible Dual

Free variables. The two constraints contradict each other ($x_1 + x_2 \geq 1$ and $x_1 + x_2 \leq -1$), and the dual equality constraints $y_1 - y_2 = 1$ and $y_1 - y_2 = 0$ also contradict.

Source: Lecture slide 55

Problem type	MIN, 2 variables, 2 constraints
Status	INFEASIBLE
Optimal Objective	–
Dual status	INFEASIBLE
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1. Model formulation

```

MIN   z = x1
subject to
    x1 + x2 >= 1      (Con1)
    -x1 - x2 >= 1     (Con2)
    x1 free, x2 free

```

Data table

Constraint	x1	x2	Sense	RHS
Con1	1	1	>=	1
Con2	-1	-1	>=	1
Objective c	1	0	min	–

Definitions

- **x1**: free variable
- **x2**: free variable

2. Result

The problem is **infeasible**. Its dual is **infeasible**. An infeasible primal has an infeasible or unbounded dual.

Dual formulation

```

MAX   z = y_Con1 + y_Con2
subject to
    y_Con1 - y_Con2 = 1      (x1)
    y_Con1 - y_Con2 = 0      (x2)
    y_Con1 >= 0, y_Con2 >= 0

```

